

Business Intelligence Implementation using Google Cloud Platform

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Business Case



XYZ E-commerce is experiencing rapid growth in online sales, leading to an increasing volume of customer data and transactions. To maintain a competitive edge, the company needs to leverage data-driven insights for strategic decision-making. However, the current business intelligence (BI) capabilities are inadequate for analyzing large datasets and providing real-time insights, which are crucial for optimizing operations and enhancing customer experiences.



Problem Statement

XYZ E-commerce faces several challenges, including:

- Scalability issues: The current system is slow and lacks the ability to handle large datasets.
- Integration difficulties: Data is scattered across multiple sources, making it difficult to integrate and analyze.
- High maintenance costs: The existing infrastructure is outdated, and the costs of maintaining and upgrading it are increasing.

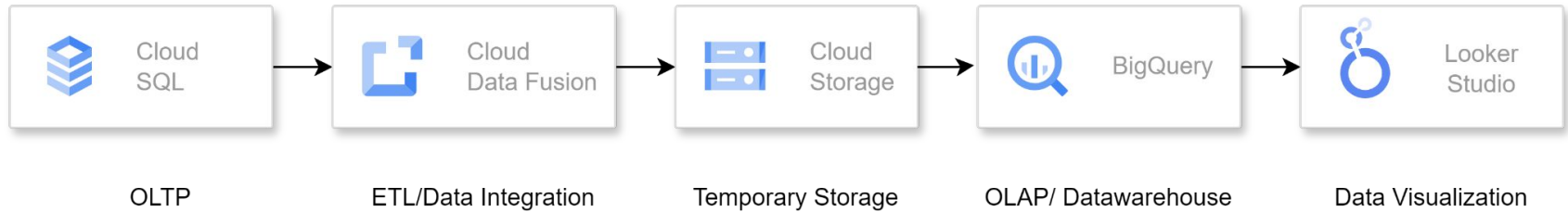


Requirements

- Cloud-Based: Scalable, flexible, and cost-effective.
- Integration: Seamless with existing systems.
- Security: Secure data storage and compliance.
- Analytics: Advanced analytics capabilities.
- Visualization: Effective data visualization tools.
- Data Handling: Large dataset processing.
- Insights: Real-time data insights.
- Collaboration: Support for team collaboration.



Suggested BI Solution & Architecture





Implementation

OLTP - Cloud SQL (PostgreSQL)

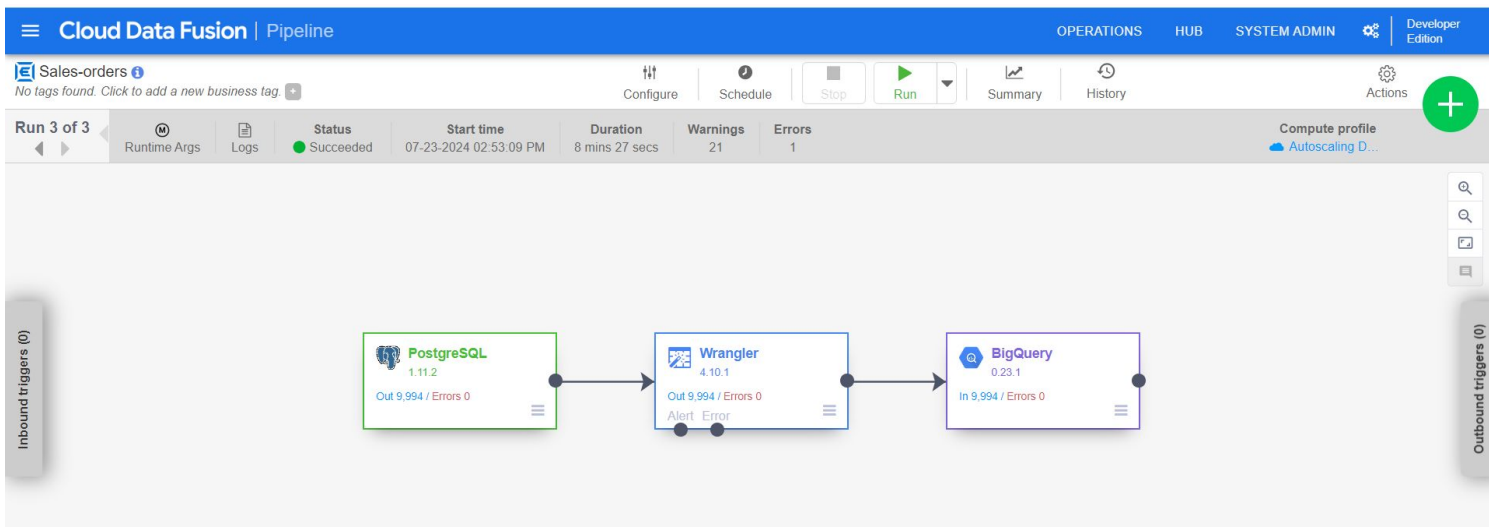
The screenshot displays the Google Cloud SQL console interface. At the top, the Google Cloud logo and 'My First Project' are visible. The main navigation bar shows 'All instances > postgre'. The Explorer panel on the left shows the database structure for 'postgres', including a 'public (Default)' schema with tables 'orders' and 'sales_orders'. The Editor panel shows a SQL query: 'SELECT * FROM public.orders'. The Results panel shows a warning that the response exceeds the size limit and a truncated table of order data.

Category	City	Country_Region	Customer_ID	Customer_Name	Manufacturer	Order_Date	Order_ID	Postal_Code	Product_ID	Product_I
Furniture	Henderson	United States	CG-12520	Claire Gute	Bush	2021-11-08...	CA-20...	42420	FUR-BO-...	Bush Soi
Furniture	Henderson	United States	CG-12520	Claire Gute	Hon	2021-11-08...	CA-20...	42420	FUR-CH-...	Hon Delk
Office Su...	Los Ange...	United States	DV-13045	Darrin Van Huff	Universal	2021-06-12...	CA-20...	90036	OFF-LA-...	Self-Adh
Furniture	Fort Laud...	United States	SO-20335	Sean O'Donnell	Bretford	2020-10-11...	US-20...	33311	FUR-TA-...	Bretford

Cloud SQL is a fully managed SQL Server database service that provides reliable and scalable OLTP (Online Transaction Processing) capabilities. It supports high availability, scalability, and security features.

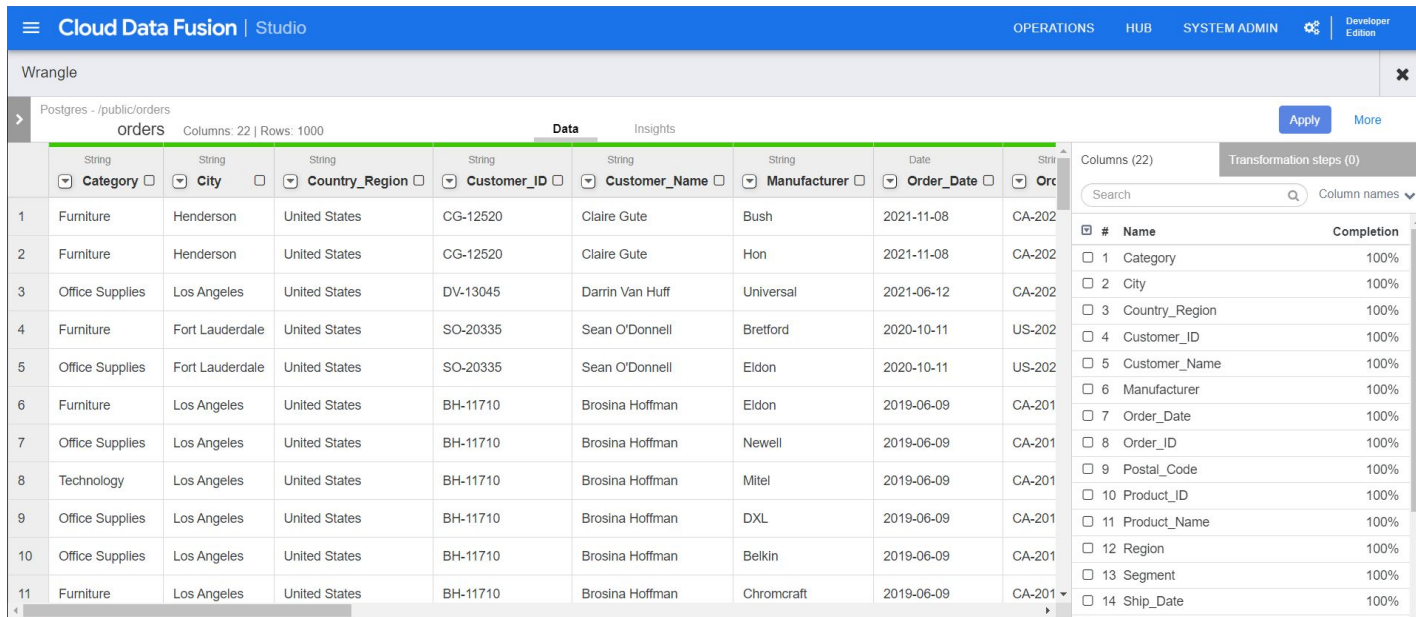
Data Integration: Cloud Fusion

Cloud Fusion is a fully managed, serverless data integration service that enables powerful ETL (Extract, Transform, Load) and data integration capabilities.



The pipeline includes data extraction from OLTP systems using Cloud SQL, data transformation with Wrangler, and loading the data into BigQuery (OLAP).

Data Integration: Cloud Fusion



Cloud Data Fusion | Studio

OPERATIONS HUB SYSTEM ADMIN

Wrangle

Postgres - /public/orders Columns: 22 | Rows: 1000

orders Data Insights

Apply More

	String Category	String City	String Country_Region	String Customer_ID	String Customer_Name	String Manufacturer	Date Order_Date	String Order_ID
1	Furniture	Henderson	United States	CG-12520	Claire Gute	Bush	2021-11-08	CA-202
2	Furniture	Henderson	United States	CG-12520	Claire Gute	Hon	2021-11-08	CA-202
3	Office Supplies	Los Angeles	United States	DV-13045	Darrin Van Huff	Universal	2021-06-12	CA-202
4	Furniture	Fort Lauderdale	United States	SO-20335	Sean O'Donnell	Bretford	2020-10-11	US-202
5	Office Supplies	Fort Lauderdale	United States	SO-20335	Sean O'Donnell	Eldon	2020-10-11	US-202
6	Furniture	Los Angeles	United States	BH-11710	Brosina Hoffman	Eldon	2019-06-09	CA-201
7	Office Supplies	Los Angeles	United States	BH-11710	Brosina Hoffman	Newell	2019-06-09	CA-201
8	Technology	Los Angeles	United States	BH-11710	Brosina Hoffman	Mitel	2019-06-09	CA-201
9	Office Supplies	Los Angeles	United States	BH-11710	Brosina Hoffman	DXL	2019-06-09	CA-201
10	Office Supplies	Los Angeles	United States	BH-11710	Brosina Hoffman	Belkin	2019-06-09	CA-201
11	Furniture	Los Angeles	United States	BH-11710	Brosina Hoffman	Chromcraft	2019-06-09	CA-201

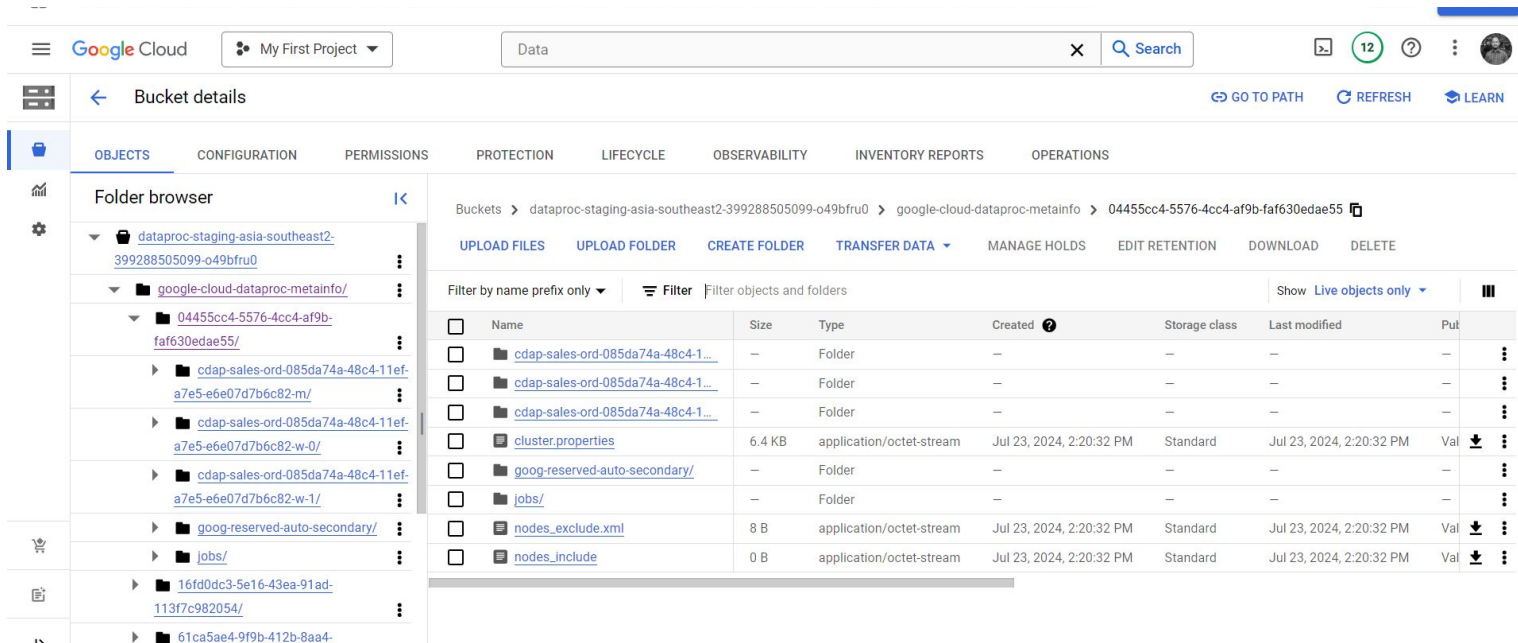
Columns (22) Transformation steps (0)

Search Column names

#	Name	Completion
1	Category	100%
2	City	100%
3	Country_Region	100%
4	Customer_ID	100%
5	Customer_Name	100%
6	Manufacturer	100%
7	Order_Date	100%
8	Order_ID	100%
9	Postal_Code	100%
10	Product_ID	100%
11	Product_Name	100%
12	Region	100%
13	Segment	100%
14	Ship_Date	100%

With data wrangling, you can transform and clean your data, as well as perform various data manipulations

Temporary Storage: Cloud Storage



The screenshot displays the Google Cloud Storage console interface. At the top, the Google Cloud logo and 'My First Project' are visible. The main navigation bar includes tabs for OBJECTS, CONFIGURATION, PERMISSIONS, PROTECTION, LIFECYCLE, OBSERVABILITY, INVENTORY REPORTS, and OPERATIONS. The 'OBJECTS' tab is active, showing the 'Folder browser' view for a specific bucket. The breadcrumb path indicates the location: Buckets > dataproc-staging-asia-southeast2-399288505099-o49bfru0 > google-cloud-dataproc-metainfo > 04455cc4-5576-4cc4-af9b-faf630edae55. The bucket name is 04455cc4-5576-4cc4-af9b-faf630edae55. The folder browser shows a tree structure with folders like cdap-sales-ord-085da74a-48c4-11ef-a7e5-e6e07d7b6c82-m/, cdap-sales-ord-085da74a-48c4-11ef-a7e5-e6e07d7b6c82-w-0/, cdap-sales-ord-085da74a-48c4-11ef-a7e5-e6e07d7b6c82-w-1/, goog-reserved-auto-secondary/, and jobs/. The main content area displays a table of objects with columns: Name, Size, Type, Created, Storage class, Last modified, and Put. The table lists several objects, including cluster.properties (6.4 KB, application/octet-stream), nodes_exclude.xml (8 B, application/octet-stream), and nodes_include (0 B, application/octet-stream).

Google Cloud

My First Project

Data

Search

Bucket details

GO TO PATH REFRESH LEARN

OBJECTS CONFIGURATION PERMISSIONS PROTECTION LIFECYCLE OBSERVABILITY INVENTORY REPORTS OPERATIONS

Folder browser

Buckets > dataproc-staging-asia-southeast2-399288505099-o49bfru0 > google-cloud-dataproc-metainfo > 04455cc4-5576-4cc4-af9b-faf630edae55

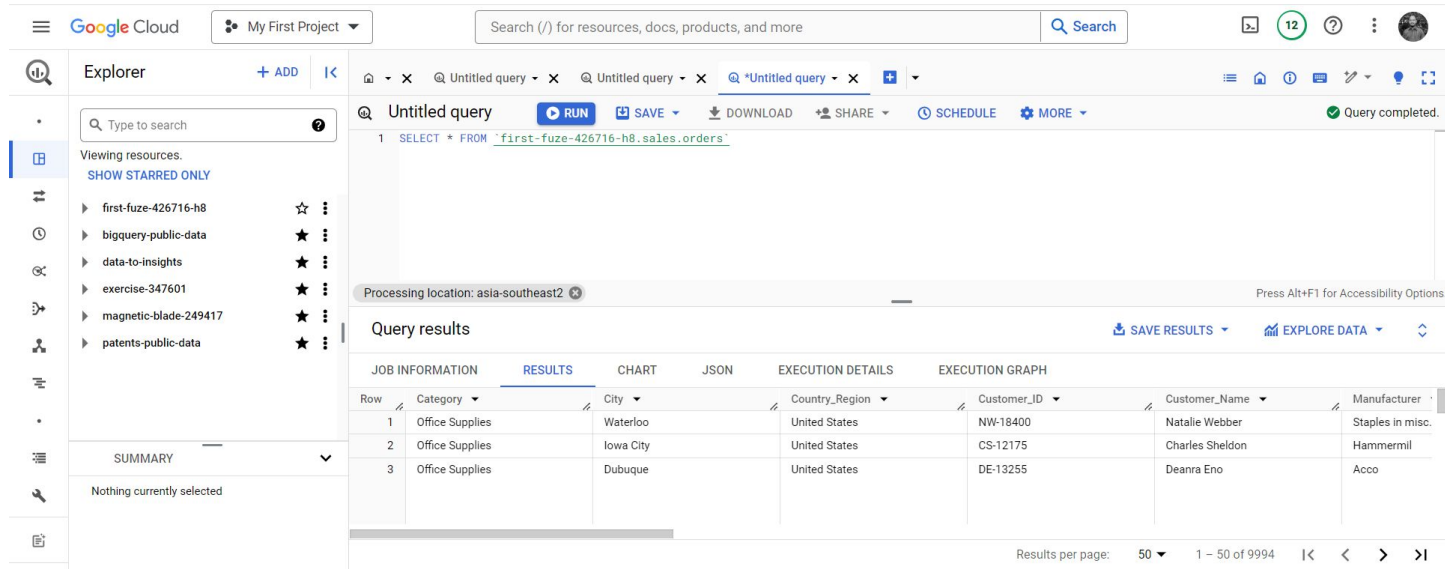
UPLOAD FILES UPLOAD FOLDER CREATE FOLDER TRANSFER DATA MANAGE HOLDS EDIT RETENTION DOWNLOAD DELETE

Filter by name prefix only Filter objects and folders Show Live objects only

	Name	Size	Type	Created	Storage class	Last modified	Put
☐	cdap-sales-ord-085da74a-48c4-1...	—	Folder	—	—	—	⋮
☐	cdap-sales-ord-085da74a-48c4-1...	—	Folder	—	—	—	⋮
☐	cdap-sales-ord-085da74a-48c4-1...	—	Folder	—	—	—	⋮
☐	cluster.properties	6.4 KB	application/octet-stream	Jul 23, 2024, 2:20:32 PM	Standard	Jul 23, 2024, 2:20:32 PM	Val ⬇ ⋮
☐	goog-reserved-auto-secondary/	—	Folder	—	—	—	⋮
☐	jobs/	—	Folder	—	—	—	⋮
☐	nodes_exclude.xml	8 B	application/octet-stream	Jul 23, 2024, 2:20:32 PM	Standard	Jul 23, 2024, 2:20:32 PM	Val ⬇ ⋮
☐	nodes_include	0 B	application/octet-stream	Jul 23, 2024, 2:20:32 PM	Standard	Jul 23, 2024, 2:20:32 PM	Val ⬇ ⋮

Cloud Fusion runs above Cloud Dataflow, which requires Google Cloud Storage for temporary storage and processing

OLAP/Data Warehouse: BigQuery



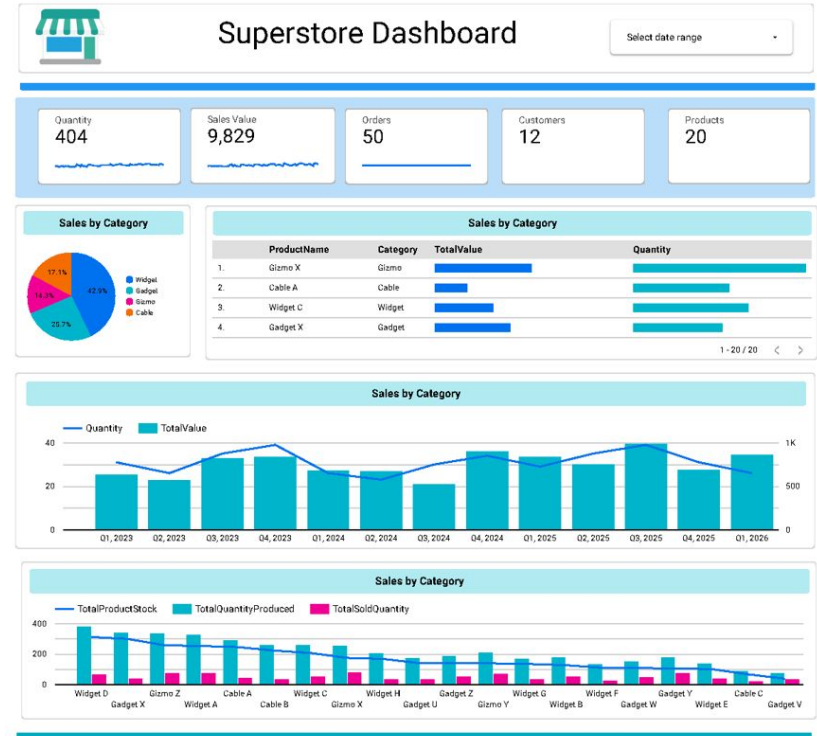
The screenshot displays the Google Cloud BigQuery console. The top navigation bar includes the Google Cloud logo, the project name 'My First Project', a search bar, and a notification badge with the number 12. The left sidebar contains the 'Explorer' panel with a search bar and a list of resources: 'first-fuze-426716-h8', 'bigquery-public-data', 'data-to-insights', 'exercise-347601', 'magnetic-blade-249417', and 'patents-public-data'. The main area shows an 'Untitled query' editor with the SQL query: `1 SELECT * FROM `first-fuze-426716-h8.sales.orders``. Below the editor, the 'Query results' section is active, displaying a table with 8 columns: 'Row', 'Category', 'City', 'Country_Region', 'Customer_ID', 'Customer_Name', and 'Manufacturer'. The table contains 3 rows of data. At the bottom, it indicates 'Results per page: 50' and '1 - 50 of 9994'.

Row	Category	City	Country_Region	Customer_ID	Customer_Name	Manufacturer
1	Office Supplies	Waterloo	United States	NW-18400	Natalie Webber	Staples in misc.
2	Office Supplies	Iowa City	United States	CS-12175	Charles Sheldon	Hammermill
3	Office Supplies	Dubuque	United States	DE-13255	Deanra Eno	Acco

BigQuery is a scalable and powerful cloud data warehouse where you can create views, tables, data marts, and schedules to meet your business requirements

Data Visualization: Looker Studio

Looker Studio is a powerful data visualization tool that transforms complex data into interactive and insightful visual reports, enabling organizations to make data-driven decisions with ease.





Pros and Cons

Pros

- Scalability: Flexible and cost-effective.
- Accessibility: Easy access to data from anywhere.
- Data Integration: Seamless integration of various data sources.

Cons

- Cost Concerns: Expenses can escalate with increased usage.
- Complex Configuration: Technical expertise required for setup.
- Internet Dependency: Limited in areas with poor connectivity.



Conclusion

The BI implementation leverages a robust architecture on Google Cloud, combining Cloud SQL for OLTP, Cloud Fusion for ETL, Cloud Storage for temporary processing, BigQuery for OLAP, and Looker Studio for powerful data visualization. This architecture ensures scalability, flexibility, and the ability to derive actionable insights, supporting informed decision-making within XYZ Company as it seeks to enhance operational efficiency and customer satisfaction.

Thank You